

MoviCare

Business & Technical Requirements Document

1. Executive Summary

MoviCare is a comprehensive mobile application designed to bridge the communication and care gap between seniors and their support network of family members and caregivers. The application addresses the growing need for accessible, real-time health monitoring and communication tools for an aging population that values independence while requiring connectivity to care providers.

Built using React Native with Expo Router and Firebase/Firestore backend, MoviCare provides a scalable, cross-platform solution that emphasizes accessibility, data security, and intuitive user experience.

2. Project Vision

To empower seniors to maintain independence and quality of life while providing family members and caregivers with peace of mind through accessible, reliable technology that facilitates health monitoring, emergency response, and seamless communication.

2.1 Target Users

- **Seniors:** Individuals aged 65+ who value independence but require health monitoring and emergency support
- **Family Members:** Adult children and relatives responsible for senior care coordination
- **Professional Caregivers:** Hired care providers requiring access to senior health information and schedules

2.2 Core Problems Addressed

- Communication barriers between seniors and their care network
- Lack of real-time health monitoring tools accessible to non-technical seniors
- Delayed emergency response due to communication challenges
- Medication non-adherence and missed appointments
- Family member anxiety about senior safety and well-being

3. Functional Requirements

3.1 Authentication & User Management

FR-AUTH-001 HIGH

Role-Based Authentication

System shall provide secure authentication with distinct user roles: Senior, Family Member, and Caregiver. Each role shall have appropriate permissions and access levels to features and data.

FR-AUTH-002 HIGH

Profile Management

Users shall be able to create and maintain profiles including personal information, emergency contacts, and relationship connections. All profile data shall persist securely in Firestore.

3.2 Health Monitoring

FR-HEALTH-001 HIGH

Health Metrics Tracking

Seniors shall be able to record and track health metrics including blood pressure (systolic/diastolic), BMI (calculated from height/weight), medical conditions, and allergies. Data shall be accessible to authorized family members and caregivers.

FR-HEALTH-002 MEDIUM

Health Data Visualization

System shall provide visual representations of health metrics over time to identify trends and patterns.

3.3 Medication Management

FR-MED-001 HIGH

Medication Scheduling

Seniors and caregivers shall be able to create medication schedules including dosage, frequency, and timing. System shall support multiple medications with varying schedules.

FR-MED-002 HIGH

Medication Reminders

System shall send push notifications to seniors at scheduled medication times. Notifications shall be clear, accessible, and dismissible.

3.4 Location & Safety

FR-LOC-001 HIGH

Emergency Alert System

Seniors shall have one-touch emergency alert capability that immediately notifies all linked family members and caregivers with the senior's current location.

FR-LOC-002 MEDIUM

Check-In System

Seniors shall be able to perform routine check-ins that share their current location and status with their care network, providing peace of mind without emergency implications.

3.5 Communication & Notifications

FR-COMM-001 HIGH

Targeted Notifications

System shall deliver notifications to appropriate recipients based on role and relationship. Seniors shall not receive family/caregiver-directed notifications.

FR-COMM-002 MEDIUM

Activity Feed

Family members and caregivers shall have access to activity feeds showing recent senior check-ins, health metrics updates, and medication adherence.

3.6 Appointment Management

FR-APPT-001 MEDIUM

Appointment Scheduling

Users shall be able to create, view, and manage medical appointments including date, time, location, and purpose.

FR-APPT-002 MEDIUM

Appointment Reminders

| System shall send timely reminders to seniors and caregivers for upcoming
| appointments.

4. Non-Functional Requirements

4.1 Performance

- Application shall load within 3 seconds on standard mobile devices
- Real-time notifications shall be delivered within 5 seconds of trigger event
- Database queries shall complete within 2 seconds under normal load

4.2 Security

- All user authentication shall use Firebase Authentication with secure token management
- Health data shall be encrypted at rest and in transit
- Access to senior data shall be strictly controlled by relationship permissions
- Location data shall only be shared during authorized check-ins or emergencies

4.3 Usability & Accessibility

- Interface shall comply with WCAG 2.1 Level AA accessibility standards
- Touch targets shall be minimum 44x44 points for senior accessibility
- Text shall be scalable and high-contrast for visually impaired users
- Critical actions shall have confirmation dialogs to prevent accidental triggers

4.4 Reliability

- System shall maintain 99% uptime excluding scheduled maintenance
- Data shall be backed up automatically in real-time via Firestore
- Application shall handle network interruptions gracefully with offline capabilities

4.5 Scalability

- Architecture shall support horizontal scaling as user base grows
- Database design shall accommodate growing historical health data
- Firebase infrastructure shall automatically scale with demand

5. Technical Stack

5.1 Frontend

- React Native with Expo SDK
- Expo Router for navigation
- TypeScript for type safety
- React Context API for state management

5.2 Backend

- Firebase Authentication for user management
- Cloud Firestore for real-time database
- Firebase Cloud Messaging for push notifications
- Expo Location for geolocation services

5.3 Development Tools

- Git for version control
- GitHub for repository management
- Expo Go for development testing
- ESLint for code quality

6. Success Metrics

- User adoption rate among target senior demographic
- Emergency alert response time under 30 seconds

- Medication adherence improvement measured by reminder completion
- Family member satisfaction with senior monitoring capabilities
- Application stability with crash rate below 1%

7. Future Enhancements

- Integration with wearable health devices (Apple Watch, Fitbit)
- AI-powered health anomaly detection
- Telemedicine video consultation features
- Multi-language support for diverse communities
- Integration with electronic health records (EHR) systems